

A Critical Survey on Real-Time Traffic Sign Recognition by Using CNN Machine Learning Algorithm

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Abstract— Real-Time Traffic Sign Recognition System (RTTSRS) is used for recognizing the traffic signboards (Take left, take right, speed limit 60 km ph... etc.), it plays a crucial role in the domains of driverless vehicles etc. By using Real-Time Traffic Sign Recognition, Traffic related problems can be reduced. It is categorized into two types- localization and recognition. Localization deals with identifying and locating traffic sign regions within the radius. Real-Time Traffic Sign Recognition is used to identify the traffic sign region within the space (rectangular) provided.

This study describes an approach for a traffic sign recognition system, many machine learning algorithms like Support Vector Machines (SVM) and Convolutional Neural Networks (CNN) have been studied for recognizing traffic signs. In this study, we have gone for critical investigation of various machine learning algorithms which gives accuracy and predict, recognize real-time traffic signs.

Keywords— Real-Time Traffic Sign Recognition System (RTTSRS), Traffic surveillance, Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Accuracy.

I. INTRODUCTION

Even though there are many machine learning algorithms, in this research paper, we will be using CNN for traffic sign recognition. There are few algorithms to solve traffic sign recognition, but CNN plays a vital role and is the more effective algorithm. Cameras will be fixed for recognition of traffic signs which will be more effective as well as useful while driving. These cameras will help in safety of the vehicle and passengers, will implement a major step on the way to independent driving. The accuracy levels will be increased or decreased concerning the implementation of the problem. For the recognition of traffic signs, image data sets will be considered for testing the required algorithm. The image data set contains images of the traffic protocols like speed limit, zebra crossing, etc. For getting accuracy, different data sets are considered, and different data sets result different accuracy. By considering more data sets, we will be able to

know which data set will be more accurate as well as effective.

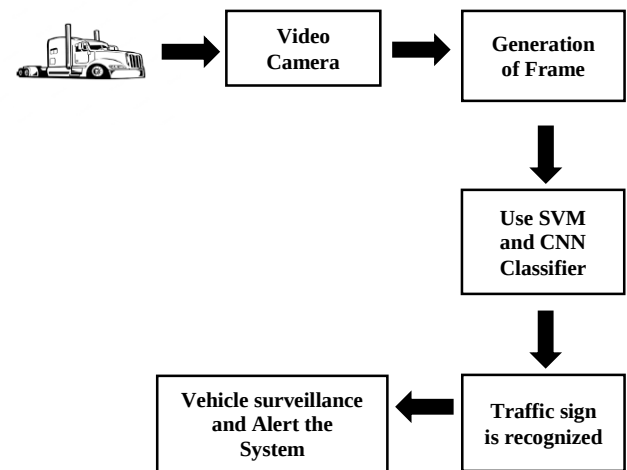


Fig.1. Pictorial representation of real-time traffic sign recognition.

Figure Labels: Pictorial representation of the traffic sign recognition system is shown in Figure 1. Figure 1 shows the lorry capturing the sign boards and pre-processing takes place. After capturing the signboard, the frame will be generated for the captured image using classifiers SVM and CNN. It finds whether it matches for dataset or not. If it matched with a dataset, the car monitor system will alert the driver.



Fig.2. Sign Boards of Direction for Specific Place.

Figure Labels: Here in Figure 2 the signboard shows the direction of a specific location named Chilakaluripet which indicates to go straight, Pathuru indicated to go in the left direction, and Kunchanapalli indicates to go in the right direction.



Fig.3. Traffic Sign Board for speed limit of 20 KMPH.

Figure Labels: In Figure 3, the traffic sign board represents the speed limit of 20 KMPH. It indicates going at 20 KMPH speed on the road. The cameras will be recognized and notifies the driver.



Fig.4. Sign Board represents a median gap.

Figure Labels: Figure 4, shows this sign indicates that there is an intersection ahead. This sign advises vehicles to slow down and watch both sides when crossing the intersection. This sign indicates that there is a gap in the road division and a U-turn is planned.



Fig.5. Sign Board of Direction for a specific location

Figure Labels: Figure 5 shows the signboard of direction for the specific location named as indicates for Hyderabad to go straight road, indicates for Market Road to take left.

II. LITERATURE REVIEW

Distributed Ensemble Learning (DEL) is an algorithm used for traffic sign recognition. DEL reduces the complexity of the data and gives accurate results. While training each model, DEL also reduces the workload.

TSR System has been split into two parts. Those are Acquisition and recognition. The recognition part consists of extracting the data from images with the help of various techniques, and the recognition which deals to recognize the traffic signs. The learning rate which indicates the weights are getting refurbished during the training. This is considered as the most important hyper parameter to tune. If the learning rate is tiny, then training that model consumes a very lengthy duration, and if the learning rate was huge, the model then intersects fast and becomes a sub-optimal solution. So, the choice of learning rate is trivial [1].

Traffic sign can be detected in two phases, i.e., detection and recognition. In detection phases, camera captures environmental conditions and detects the presence of sign in the frame. In recognition phase, algorithms classify the detected sign with the help of TensorFlow, machine learning framework etc. Multi detection process is used by author which involves color and shape detection which helps to pass the recognition phase with better accuracy. Training time of a model is reduced a lot by using GPU and training of 3CNNs simultaneously.

As the model depends on image, if there is any kind of reflection or if the signboard gets cut-off or during nighttime, cameras fail to detect the sign during nighttime due to darkness, the sign can't be detected and classified. So, the author suggested a text to speech module for the model [4].

Intelligent transportation can help reduce the number of daily traffic accidents around the world. The authors worked on intelligent transportation models which uses video processing system in a big data environment. This model uses machine learning and neural network algorithms to recognize traffic signs from videos in big data environment. The performance of this system is tested under various conditions and environments, and detection efficiency and accuracy are calculated. Finally, the authors were able to recognize traffic signs with high practical application value of the system's design scheme [9].

Traffic sign recognition plays an important role driving and driver assistance systems. To improve the real-time detection of small targets in YOLOv4, the author used his K-Means to cluster tags and improve small tag location accuracy. After experimentation and testing, we found that the improved algorithm improves the accuracy of real-time traffic sign recognition and performs better in terms of performance and speed compared to the original YOLOv4 [2].

Traffic sign recognition plays an important role in intelligent driving systems like auto driving. Gradually the research field on neural networks has become quite popular and target classification effect and capabilities are getting better day by day. The proposed CNN model was tested on GTSRB (German traffic sign recognition benchmark) and compared with other researcher's models. Data Enhancement and process is done to classify the data accurately by enhancing quality. The model is experimented and tested with GTSRB dataset which includes over 50000+ images in 40+ categories from 15x15 to 250x250 dynamically changed resolution. After calculating and verifying the test results against GTSRB dataset author proposed. Processing time, simple architecture and strong scalability are some of the pros of the proposed model. Recognizing traffic signs under critical weather and tested against multiple benchmark datasets are proposed as future research by the author [13].

A real time sign recognition model for traffic sign detection was proposed by using machine learning to help the safety of travelers by preventing accidents with warning. It extracts and process a traffic sign from complex and natural images and alerts the driver. It helps drivers by alerting through sounds and helps to make fast decisions. The proposed

architecture is sectioned into three phases-pre-processing, determining input and resize the information. Before, pre-processing noise is removed from the data to reduce complexity and improve the precision. A camera is attached to the car and used to take pictures. In the second and third phases, convolutional neural networks (CNN) are used. The authors clearly categorized all the advantages and limitations of the algorithm and proposed the traffic sign detection and recognition achieved by the CNN algorithm. The proposed system issues voice alerts when traffic signs are identified [14].

In this article, the authors proposed a design and implemented a detector based on R-CNN (Convolutional Neural Network). This design use shape and color info to pinpoint the location of small signs. A powerful CNN with an asymmetric kernel is used as classifier. A public benchmark dataset is used to test both the detector and the classifier. The authors clearly specified the network architecture, optimization, and regularization. Finally, it is shown that the proposed detector can recognize traffic signs beyond the limits of color and shape, and the proposed classifier is proved to be simple and powerful according to the evaluation results. The author said further future work would improve performance and improve robustness [15].

III. CAMERAS USED FOR RECOGNIZING TRAFFIC SIGN

For traffic sign recognition, a high-definition camera is used. These high-definition cameras are more effective at recognizing signs. They are mostly used for complicated traffic signs. For capturing high-res images of traffic symbols and signs, an active bifocal camera system has been proposed. Wide-angle cameras are supported by active telephoto cameras. These two cameras are used to capture high-definition images of road signs, as well as the shapes and colors of the signs. The two cameras are linked through traffic sign recognition. As a result, a dual focal active camera performs better when mounted in a car and driving.

IV. ALGORITHMS USED FOR RECOGNIZING TRAFFIC SIGNS

According to research results, the following algorithms are used for recognizing traffic signs are Convolutional Neural Networks (CNN), Regions of Convolutional Neural Networks (R-CNN), Fast R-CNN, Faster R-CNN, Distributed Ensemble Learning (DEL), Stochastic Gradient Descent Algorithm, YOLOv3, YOLOv4, CNN Ensemble, Support Vector Machine (SVM), etc. For getting a more accurate value CNN is the best algorithm. CNN is not more complex. CNN does not require human supervision for the task of identifying key features. They are highly accurate in image recognition and classification.

TABLE I. ACCURACY OF REAL-TIME TRAFFIC SIGN RECOGNITION

Paper	Recognition Process	Accuracy
[1]	DEL	93%
[3]	YOLO v3	92.2%
[4]	CNN	99.94%
[4]	SVM	98%
[7]	YOLO v4	81.7

V. ALGORITHMS USED FOR TRAFFIC SIGN RECOGNITION

Though R-CNN and Fast R-CNN are the advanced modes of CNN in traffic sign recognition CNN gives more accuracy than any other so we can conclude that CNN gives more accuracy than any other algorithm.

A. Convolutions Neural Network(CNN)

Convolutions Neural Network (CNN) algorithm works more efficiently, and it is more effective. It is a traffic sign recognition and recognition method that uses image processing to recognize signs and a collection of convolutional neural networks (CNNs) to recognize signs. CNN has a high recognition rate. The disadvantage lots of training data is required [4].

B. Domain with Convolutional Neural Networks (R-CNN)

A domain with a convolutional neural network (R-CNN) connects the sentences in the domain whose features are associated with the convolutional neural network (CNN). This algorithm is a multilevel detection algorithm. This algorithm is the Multistage detection algorithm. R-CNN is an extension of CNN with an object recognition. For image classification and image recognition CNNs are used. Because CNN works some more efficiently compared to R-CNN [15].

C. Distrubuted Ensemble Learning (DEL)

Distributed Ensemble Learning (DEL), an effective way to automatize the traffic sign recognition. The three main goals of distributed ensemble learning are reducing complexity, reducing the training load for each model, and increasing convergence. In this algorithm, deep learning and distributed learning will be placed. In this method, accuracy will be classified more accurately. Distributed Ensemble learning reduces both the complexity and training time while increasing accuracy. Depending on the number of employees, the length of the training is shortened. It also reduces the complexity of the number of workers and training periods [1].

D. Yolov3

In the field of autonomous vehicles, real-time traffic sign recognition technology plays a critical part in automotive assistance systems and is frequently used. The continuous development of various types of target/image recognition technologies based on deep learning technology will play an important role in self-driving automobiles. Computer hardware and technology have undergone significant changes in recent years. YOLO has been updated to numerous versions, including YOLO v2, v3, and v4, depending on the

requirements and precision of target object identification. The YOLO V3 model's network structure for Darknet-53 continues to use its YOLO V3 forecasts on its three scales. This is precisely provided by resampling the input image's dimensions by 32, 16, and 8 correspondingly. The YOLO v3 version is better at identifying small things [3].

E. Yolov4

To identify target objects along the entire route, YOLO techniques were applied. The target identification mechanism of YOLO v4 is an improved algorithm for location accuracy that uses K-Means clustering. Based on object recognition of YOLO v3 algorithm, data processing optimization, backbone network, network training, activation function and loss function provide the best detection/recognition speed and accuracy, YOLO v4 uses backbone network to extract input image features.

The primary network structure of YOLO v4 is CSPDark Net53, and two algorithms—Pan Net and SPP structure—are used to enhance YOLO v4. Currently, the YOLOv4 is the most advanced model in the YOLO. However, YOLOv4 is still not optimized for all cases, which means that when the dataset contains many small objects, YOLOv4 is still not accurate for cases such as autonomous driving scenarios where there are many small and distant objects on the road, such as pedestrians, road signs, etc.

F. support vector machine (svm)

Identifying the ideal hyper-plane that can divide class borders as widely as possible. The ratio of the number of elements to the hyperplane's size is direct. When there are numerous functions, it is challenging to visualize the hyper-plane. In real-time traffic sign recognition, we employ a support vector machine (SVM) to classify road sign forms and symbols. We use linear, polynomial, RBF, and sigmoid kernel functions in SVM classification, and the linear function performs better than the other types. After receiving image data from the camera, SVM plots them on the graph based on patterns and differentiates the road sign forms and symbols by the optimal hyperplane

VI. IMAGE DATA

Image data is used for traffic sign recognition. Image data includes images of traffic signs, speed limits, crosswalks, etc. Images are captured with a high-resolution camera in Dimensions of 25 x 15 inch and Size of the image will be 4500 x 3000 pixels or 12MP. The camera used to capture the character is an active dual-focus camera. An active dual-focus camera is a high-resolution camera that captures images at a higher resolution. Photos from the camera are in perfect shape. The camera can recognize color markings on traffic signs. A dual-focus camera system can capture high-resolution images of road signs from a greater distance than conventional single-camera systems. This camera recognizes rapidly and accurately. Along with the wide-angle camera, it also contains an active telephoto lens. These cameras can assist drivers and increase traffic safety. Similarly, there are numerous cameras used to capture traffic signs in high definition. Dual focal active cameras are more effective at

recognizing traffic signs here. It is also equipped with an active tele camera for traffic sign recognition.



Fig.6. Using Keras and deep learning, the German GTSRB (German Traffic Sign Recognition Benchmark) dataset are used to classify traffic signs.

Source: <https://steemit.com/programming/@kasperfred/looking-at-german-traffic-signs>

VII. DATA SETS

The dataset which we have implemented is Malaysian Dataset. The initial data set is split into test and train datasets with the corresponding ratios (approx. 2:1) in order to evaluate the model. At the training stage, it has processed 46502 images of for a single iteration. And at testing stage 25037 images are processed. Now from all the 100 iterations, intermediate accuracy will be calculated with group of images which were processed at the training stage. After training, the accuracy is calculated from the test data set. By calculating the data set, we will be able to know how the accuracy levels are decreased and increased. We will be able to know the accuracy growth with the increasing or decreasing the number of training iterations.



Fig.6.1. Different Kinds of Sign Boards

Source: <https://www.godigit.com/traffic-rules/traffic-signs-in-india>

Figure Labels: The above image describes about the various types of sign boards. Each sign board represents their specific traffic indications.



Fig.6.1. Dataset of Sign Boards

Source: <https://paperswithcode.com/task/traffic-sign-recognition>

Figure Labels: The above figure represents model dataset that we will be using in our real time traffic sign recognition system project (RTTSRS).

VIII.CONCLUSION

In this paper, we conclude that CNN is the best method for the recognition of traffic signs after testing it out with many datasets containing traffic sign images, comparing and analyzing the accuracy of different types of traffic sign recognition processes and algorithms like DEL, YOLO v3, YOLO v4, CNN, Support Vector Machine (SVM), etc.,

According to the evaluation results CNN performs with better and high accuracy compared to other sign recognition processes. CNN detects objects quickly and requires little pre-processing; it automatically detects important features without human intervention, and CNN is not more complex.

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